Lesson 23 Pigeonhole Principle

1. Suppose N is an integer at least ten billion, prove that N has at least two same digits. What is the largest integer with no repeated digits?
2. Find the smallest number of people such that there will certainly be two with the same birthday.
3. Find the smallest number of people such that there will certainly be three with the same birthday.
4. Prove that among six random integers, two will have the same remainder when divided by 5.
5. Prove that among any seven integers, the sum or difference of two of them is a multiple of 10.

   Hint: Partition the integers into 6 categories by their remainders when divided by 10, r or $10 - r$:
   - 0
   - 1 or 9
   - 2 or 8
   - 3 or 7
   - 4 or 6
   - 5
6. A long shelf on a math professor's library wall holds 28 books, all of them in the fields of algebra, calculus, or discrete math. Prove that there must be at least 10 books in one of these 3 areas.